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Message from the Conference Chairs

Dear friends and colleagues,

Welcome to the 39th ACM SIGSIM International Conference on Principles of Advanced Discrete Simulation (PADS) in Santa Fe, New Mexico. For several decades, ACM SIGSIM PADS has been an international forum for presenting and sharing recent research results and propelling methodological advances in modeling and simulation. This year's edition will not be different: PADS'25 features a lively, engaging, and stimulating program with many opportunities for discussing new results, ongoing research projects, and new challenges and opportunities.

We thank:

- The Technical Program Committee, consisting of 41 distinguished experts, plus the external reviewers,
- The session chairs who solicited papers in traditional and emerging fields of modeling and simulation,
- The chairs of the PhD colloquium,
- The chair of reproducibility and artifact evaluation,
- The local chair, and,
- The PADS steering committee.

Their combined, invaluable efforts made this year's excellent program possible.

We particularly thank the keynote speakers of this year: J. Stephen Lansing (Santa Fe Institute) and Vibeke Sørensen (Complexity Science Hub Vienna). They will jointly present a keynote on "Phase transitions in Balinese time, music and irrigation management". We are looking forward to a thought-provoking talk and discussions.

As before, the conference is organized in thematic sessions that reflect the diversity of topics of relevance for discrete event modeling and simulation. For the second time, we had two separate calls for papers. To the first call, 24 papers, and for the second call, 28 papers (which included seven revised papers of the first call) have been submitted. All papers were reviewed by at least three members of the program committee. Within a virtual program committee meeting, each paper and its reviews were discussed. Four papers have been accepted for the ACM Transactions on Modeling and Computer Simulation (TOMACS) PADS'25 special issue, and 12 full and eight short papers for the ACM PADS'25 Proceedings. Without short papers, this results in a 35% acceptance rate.

In addition, the ACM PADS'25 Proceedings contain four extended abstracts

presenting a synopsis of the papers that are published in the TOMACS PADS'25 special issue, four extended abstracts about work to be presented at the PhD colloquium, eight reproducibility reports about papers published in the proceedings, a summary of the keynote given by researchers associated with the Santa Fe Institute, and reflections on the two PADS Test of Time awarded in 2025.

The recipients of the Test of Time Awards for publications in 2005 and 2015, respectively, are:

- Michael Liljenstam, Jason Liu, David Nicol, Yougu Yuan, Guanhua Yan, and Chris Grier. 2005. RINSE: The Real-Time Immersive Network Simulation Environment for Network Security Exercises. In Proceedings of the 19th Workshop on Principles of Advanced and Distributed Simulation (PADS '05). IEEE Computer Society, USA, 119–128.
- Davide Cingolani, Alessandro Pellegrini, and Francesco Quaglia. 2015. Transparently Mixing Undo Logs and Software Reversibility for State Recovery in Optimistic PDES. In Proceedings of the ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (SIGSIM PADS '15). ACM, USA, 211–222.

The candidates short-listed for the PADS'25 Best Paper Award for this year (in no particular order) are:

- Xin Wang, Kevin Brown, Robert Ross, Christopher Carothers and Zhiling Lan: MFNetSim: A Multi-Fidelity Network Simulation Framework for Multi-Traffic Modeling of Dragonfly Systems.
- Bo Zhang, Wen Jun Tan, Wentong Cai, and Allan Zhang: Privacy Meets Performance: Enhancing Distributed Simulation-based Federated Multi-agent Learning with Privacy-preserving Surrogate Model.
- Artem Konev, Bernhard Sadransky, Silvana Rauer-Zechmeister, Daniel Cornel, Ingo Schwerdorf, Jürgen Waser, and Milena Vuckovic: Interactive geospatial stories for flood management.
- Justin N. Kreikemeyer, Miłosz Jankowski, Pia Wilsdorf and Adelinde Uhrmacher: Using (Not-so) Large Language Models to Generate Simulation Models in a Formal DSL: A Study on Reaction Networks.

The best paper award winner from the candidates above will be announced during the PADS'25 banquet. In keeping with tradition, we announce the PADS'24 Best Paper Award:

- Srikanth Yoginath, Pratishtha Shukla, Sudip Seal and James Nutaro. SpecSims: A Scalable Speculative Tree-based Simulation Cloning Framework For Finite Memory Machines.

PADS participates in ACM's Artifact Evaluation initiative, strengthening the trust in published modeling and simulation research. Overall, artifacts for 10

of the accepted papers were submitted for evaluation. After careful evaluation, the reproducibility committee could award badges to all papers that opted into the initiative. We sincerely thank the authors and the reproducibility committee for making this valuable contribution to the community.

The success of a conference is mainly on the participants' part; we warmly thank all contributors and participants. We wish you a productive, engaging conference in Santa Fe that sparks new ideas and collaborations.

June 23rd, 2025, Santa Fe, NM, USA

Ernest Page
SIGSIM PADS '25 General Chair
The MITRE Corporation

Adelinde M. Uhrmacher
SIGSIM PADS '25 Program Chair
University of Rostock

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